

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.35	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.9	Product Info 2	A1
Occupation	Software Developer	Annual Income	\$120,000.00
Insurance Type	Life	Coverage Amount	\$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant Details:

* **Age:** 0.35 years

* **Height:** 175.0 cm

* **Weight:** 70.0 kg

* **BMI:** 22.9

* **Occupation:** Software Developer

* **Income:** \$120,000.0

* **Smoker:** 0 (Non-smoker)

* **Previous Claims:** 0

* **Family History:** 0 (No adverse history)

* **Insurance Type:** Life

* **Coverage Amount:** \$500,000.0

* **Exercise:** 1 (Regular)

* **Alcohol:** 0 (None)

* **Gender:** Male

* **ML Predicted Risk Class:** 1 (out of 8)

* **ML Prediction Confidence:** 60.3%

* **Calculated Annual Premium:** \$10,000.00

1. Risk Assessment

The applicant presents a profile that, on the surface, indicates low actuarial risk across most health and lifestyle metrics.

* **Health & Lifestyle:**

* **BMI (22.9):** Falls within the healthy range, indicating good physical condition.

* **Smoker (0), Alcohol (0), Exercise (1):** These are very favorable factors, suggesting a healthy lifestyle with reduced health risks.

* **Previous Claims (0), Family History (0):** Absence of adverse claims history and family medical conditions further reduces risk.

* **Occupation & Income:**

* **Occupation (Software Developer):** Generally considered a low-risk, sedentary profession.

* **Income (\$120,000):** A strong income level, which supports the proposed coverage amount in a typical adult context.

* **Age:** The reported age of 0.35 years (approximately 4 months) is highly unusual for a standard life insurance application, particularly when combined with an occupation and income. This is a critical anomaly that requires immediate clarification and casts doubt on the validity of the application data as a whole.

* **Coverage Amount (\$500,000):** For a typical adult with the reported income, this coverage amount (approx. 4.17x income) would be reasonable. However, for an infant, this amount is exceptionally high and inappropriate for a standard life insurance policy, further highlighting the age discrepancy.

2. Comparative Analysis

The machine learning classifier has assigned a Risk Class of 1 and a Calculated Annual Premium of \$10,000.00. This aligns perfectly with two of the top similar historical approved cases:

* **POL-33998 (Risk Class: 1, Premium: \$10,000.00, Status: approved)**

* **POL-49239 (Risk Class: 1, Premium: \$10,000.00, Status: approved)**

The third similar case, POL-8902, was Risk Class 2 with a slightly higher premium of \$12,000.00, suggesting the current applicant is assessed at a marginally lower risk profile than POL-8902, which is consistent with a Class 1 assignment.

The "100.0% Similarity" indicates that, based on the input features used by the model, the current application is deemed identical to these historical cases in terms of risk profile. If the applicant's data (excluding the age anomaly) is accurate, the ML model's output and premium calculation are well-supported by these precedents.

3. Anomaly & Fraud Detection

A significant anomaly is present in this application:

* **Age vs. Occupation/Income:** The applicant's age is stated as 0.35 years (approximately 4 months old), yet the occupation is "Software Developer" with an income of \$120,000.00. This is a fundamental contradiction and a clear data inconsistency. An infant cannot hold such an occupation or earn this level of income.

* **Coverage Amount for Age:** The requested coverage of \$500,000.00 is highly excessive and inappropriate for a 0.35-year-old child under a standard life insurance policy.

This discrepancy strongly suggests either:

1. **Data Entry Error:** The most probable explanation, where the age field may have been incorrectly populated (e.g., '0.35' instead of '35' or another adult age).
2. **Attempted Fraud/Misrepresentation:** Less likely given the direct contradiction, but it cannot be entirely ruled out without further verification.

There are no other obvious anomalies in the provided data (e.g., unusual BMI, high claims history, inconsistent lifestyle indicators *if assuming an adult applicant*). The primary concern stems from the age-occupation-income conflict.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 for a Risk Class 1 applicant is consistent with the output of the machine learning model and is directly supported by two of the three most similar historical approved cases.

* **Risk Class 1:** This indicates a very low-risk profile based on the reported health, lifestyle, and occupation factors (non-smoker, healthy BMI, regular exercise, low-risk job).

* **Historical Precedent:** The exact match with POL-33998 and POL-49239 provides strong justification for this premium level, assuming the underlying risk factors are accurately represented.

* **Prediction Confidence:** The 60.3% confidence level, while not extremely high, suggests a reasonable

degree of certainty from the model in its classification.

However, **this premium recommendation is contingent on the applicant's core demographic data being accurate.** Due to the severe age-occupation-income anomaly, the validity of the premium calculation is currently undermined. If the applicant is genuinely an adult matching the other positive risk factors, the \$10,000.00 premium appears appropriate. If the age of 0.35 is correct, this premium (and the entire application) would be nonsensical.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review and Data Verification****

Due to the critical and irreconcilable inconsistency between the applicant's stated age (0.35 years) and their occupation ("Software Developer") and income (\$120,000), this application cannot proceed without immediate manual intervention.

The underwriting team must:

1. ****Verify Applicant's True Age:**** Confirm the correct age and date of birth of the proposed insured.
2. ****Confirm Occupation and Income:**** Re-validate the occupation and income details against the verified age.
3. ****Clarify Application Type:**** Determine if this is a primary policy for an adult, a child rider, or a specific juvenile product, as the coverage amount is highly unusual for an infant.

****Contingent Actions:****

* ****If Age is a Data Entry Error (e.g., actual age is 35):**** If the corrected data aligns with the other positive risk factors, the application may be suitable for approval at the calculated premium of \$10,000.00.

* ****If Age of 0.35 is Confirmed:**** The application, as presented, is fundamentally flawed and should be declined or redesigned as a child-specific policy with appropriate coverage limits, as an infant cannot be a "Software Developer" earning \$120,000.00. This scenario would also require investigation into potential misrepresentation.

The current application cannot be approved or rejected based solely on the ML output without resolving this critical data integrity issue.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.